Univalent Bordism Categories are Controlled

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1 Introduction

Here is (up to potential mistakes) how the usual definition of the category of cobordisms goes:

Definition 1.1 (Pre-Bordism Category)**.** We define an $(n+1)$-fold simplicial set $\mathcal{P}\mathrm{Bord}^n$ as follows. The set $\mathcal{P}\mathrm{Bord}^n_{\mathbf{i},j}$ is given by the set of

- Smooth maps $\left((a^l_k, b^l_k \in \Delta^j_e \to \mathbb{R})_{0\leq k\leq i_l}\right)_{1\leq l\leq n}$ such that

$$\begin{cases} a^l_k < b^l_k \\ a^l_{k-1} \leq a^l_k \\ b^l_{k-1} \leq b^l_k \end{cases}$$

- M compact $(n+j)$-dimensional submanifold of $\mathbb{R}^\infty \times \left(\prod_{1\leq l\leq n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e$ such that:
  (a) **Family:** the composition $\pi\colon M \hookrightarrow \mathbb{R}^\infty \times \left(\prod_{1\leq l\leq n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e \twoheadrightarrow \left(\prod_{1\leq l\leq n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e$ is proper, and the further projection to Δ^j_e is a submersion which is trivial on the extended part of the simplex,
  (b) **Reducedness:** for every $1 \leq d \leq n$ and every $0 \leq k \leq i_d$, $M_| \twoheadrightarrow \left((a^d_k, b^d_k) \times \prod_{d\leq l\leq n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e$ is a submersion,

Famously, this is not univalent. The goal of this paper is to construct its univalent completion explicitly.

1.1 Tameness

Definition 1.2 (End Space)**.** Let W be a topologial space. The **end space** $e(W)$ of W is the space of **proper paths** $[0,\infty) \to W$.

Definition 1.3 (Ends)**.** Let W be a topologial space. A neighbourhood of infinity in W is an open subset $U \subset W$ that contains the complenent of a compact. An **end** is an equivalence class of decreasing sequences of neighbourhood of infinity $\ldots \subset U_n \subset \ldots \subset U_1 \subset W$ such that $\bigcap_{n\geq 1} \overline{U_n} = \emptyset$. They're subject to the obvious equivalence relation.

Definition 1.4 (Tame Ends)**.** And end is **tame** if the inclusions $U_i \hookrightarrow U_j$ are homotopy equivalences for all $i < j$.

1.2 Control Topology in Families

Definition 1.5 (Controlled Famillies, I)**.** Let X be a smooth manifold and I a compact manifold with boundary. Let $W \subset \mathbb{R}^\infty \times I^o \times X$ be a submanifold such that

- $W \to X$ is a smooth fiber bundle,
- for all $t \in X$, $W_t \twoheadrightarrow I$ is a proper submersion,

We say that $W \to I \times X$ is **controlled** in the I-direction if it extends to a topological subspace $\overline{W} \subset \mathbb{R}^\infty \times I \times X$ such that $\overline{W}^o = W$ and for all $t \in X$, the projection $\partial\overline{W}_t \simeq \partial I$ is a homeomorphism.

I think controlled topology should give ways to re-write this. Firslty, it should be equivalent to $O(1)$-variance. Secondly, for sufficiently high dimensions, this should become an algebraic condition. This second aspect might help proving that the space of objects of $\mathcal{B}\mathrm{ord}^n$ is contractible in the PL-case.

Pablo: formulate a precise result

Definition 1.6 (Controlled Famillies, II)**.** Some definition using $O(1)$-variance.

1.3 Univalence is Control

Definition 1.7. We define an $(n+1)$-fold simplicial set $\mathcal{B}\mathrm{ord}^n$ as follows. The set $\mathcal{B}\mathrm{ord}^n_{\mathbf{i},j}$ is given by the set of

- Continuous maps $\left((a^l_k, b^l_k \in \Delta^j_e \to \mathbb{R})_{0\le k\le i_l}\right)_{1\le l\le n}$ such that

$$\begin{cases} a^l_k < b^l_k \\ a^l_{k-1} \le a^l_k \\ b^l_{k-1} \le b^l_k \end{cases}$$

- M compact $(n+j)$-dimensional submanifold of $\mathbb{R}^\infty \times \left(\prod_{1\le l\le n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e$ such that:
  (a) **Family:** the composition $\pi\colon M \hookrightarrow \mathbb{R}^\infty \times \left(\prod_{1\le l\le n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e \twoheadrightarrow \Delta^j_e$ is a smooth fiber bundle,
  (b) **Reducedness:** for every $t \in \Delta^j_e$, $1 \le d \le n$ and every $0 \le k \le i_d$, $M_t \twoheadrightarrow (a^d_k(t), b^d_k(t)) \times \prod_{d\le l\le n}(a^l_0(t), b^l_{i_l}(t))$ is a proper submersion of codimension $d-1$,
  (c) **Control:** the t-families above of proper submersions are controlled in the categorical directions.

Remark 1.1 (Inner Control is Guaranteed). The control property should only be necessary for $k = 0$ and $k = i_d$. For an "inner" k, I expect that one can quotient out the rest of W to obtain the control extension. More precisely, suppose one has $a_0 < a_k$ and $b_k < b_{i_d}$. Then one can consider the quotient of $\mathbb{R}^\infty \times \left(\prod_{1\le l\le n}(a^l_0, b^l_{i_l})\right) \underline{\times} \Delta^j_e$ by ...

Pablo: to be formalized

Remark 1.2 (Continuity vs Smoothness). It seems to me like the a, b maps only need to be continuous. It would be interesting to see if more regularity arises in my constructions.

Remark 1.3 (Looping). One can replace (b) by "proper submersion of codimension d" and this becomes $\Omega^d\,\mathcal{B}\mathrm{ord}^n$ where objects are compact d dimensional manifolds. It's sufficient to treat the $d = 0$ case as it contains all other cases through this observation.

Theorem 1.1

$\mathcal{B}\mathrm{ord}^n$ is the univalent completion of $\mathcal{P}\mathrm{Bord}^n$.

1.4 Quinn's Theorem

I think Quinn's theorem should show how to compute the homotopy type of the object space of $\mathcal{B}\mathrm{ord}^n$ in terms of controlled topology. In particular, existence of PL-manifold structures on mapping cylinders is eactly what is needed to then use Alexander's trick and show contractibility.

2 Univalence

Proposition 2.1

Let $X \to Y$ be a morphism of n-fold Segal spaces. Then it is a univalent equivalence if and only if:

- *$X^\simeq \to Y^\simeq$ is a weak homotopy equivalence of spaces*
- *for $1 \le k \le n$, we have homotopy pullbacks*

$$\begin{tikzcd} X_k \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & Y_k \arrow[d] \\ {[\partial c_k, X]} \arrow[r] & {[\partial c_k, Y]} \end{tikzcd} \tag{2.1}$$

This should follow from "ITERATED SPANS AND CLASSICAL TOPOLOGICAL FIELD THEORIES" by Haugseng (section 7) and from Lurie's work on "distributors" in "$(\infty, 2)$-Categories and the Goodwillie Calculus I". Easy inductive argument.

Note: one also has John's formula for the completion with Θ-spaces, if necessary (which generalizes the OG formula from Rezk). The advantage of this formula is that (maybe) it makes it so one only has to do the zig-zag thing once. But could be very difficult to actually work out the homotopy colimits.

3 Proof of the Main Theorem

Remark 3.1 (Proof Strategy). The proof will be divided in two main steps:

- Show Univalence: this should involve construction the zig-zag with the invertible data, and show this already appears as a controlled object.
- use Proposition 2.1 to show it's a completion. This involves:
  (a) tbd
  (b) second part should be easier?

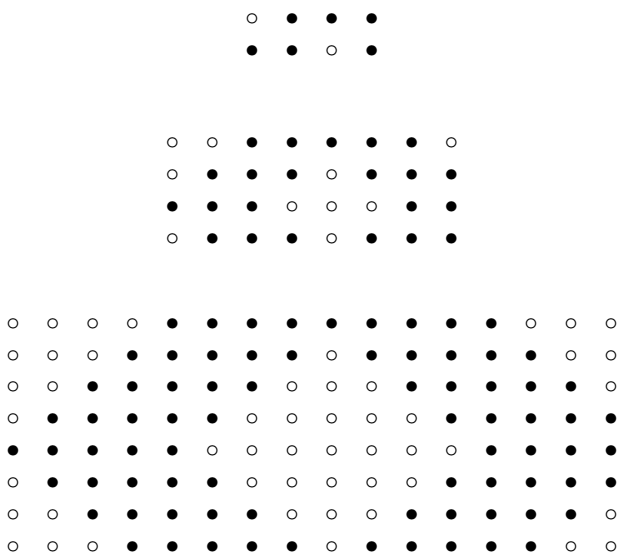
3.1 Invertible Data as a Paving

For $n \geq 0$, we have a distance on $\mathbb{Z}/2^n\mathbb{Z}$ given by the cyclic picture: $d(i,j) = \min\{|i-j|, 2^n - |i-j|\}$ for all $i, j \in \mathbb{Z}/2^n\mathbb{Z}$. This is the shortest path distance for the undirected graph given by the edges $(i, i+1)$ for all $i \in \mathbb{Z}/2^n\mathbb{Z}$. This induces a product distance on $\mathbb{T}_{2^n} := \mathbb{Z}/2^n\mathbb{Z} \times \mathbb{Z}/2^{n+1}\mathbb{Z}$.

Definition 3.1 (Exclusion on the Modulo Torus)**.** For $n \geq 0$, we define a subset $E_n \subset \mathbb{T}_{2^n}$ as

$$E_n = \{(i,j) \in \mathbb{T}_{2^n} \mid (i,j) \text{ is at distance } \geq 2^{n-1} \text{ from both } (0,0), (2^{n-1}, 2^n)\}$$

Bellow is an illustration of the first few E_n:



Definition 3.2 (Nodes of the Paving)**.** For $n \geq 0$, we define a subset $A_n \subset \mathbb{Z}^n$ as

$$A_n = \{i \in \mathbb{Z}^n \mid \forall 0 \leq j \leq n-1,\ (i_j, i_{j+1}) \in E_n\}$$

Definition 3.3 (Paving)**.** A rectangle in $\mathbb{Z}^n$ is a choice of a set of corners $R = \prod_{i=1}^n R_i \subset \mathbb{Z}^n$ where $R_i \subset \mathbb{Z}$ is of cardinality either 1 or 2. A paving is a collection of rectangles forming a cell decomposition of $\mathbb{R}^n$. For $n \geq 0$, we have a paving $\mathcal{A}_n$ of $\mathbb{R}^n$ given by the following rule:

- for all rectangles R, we have $R \in \mathcal{A}_n$ if and only if $R \subset A_n$ and $\mathsf{Hull}(R)^\circ \cap A_n = \emptyset$.

Remark 3.2 (Wavelets). Consider $f = \sum_{i=1}^n \cos(\pi x_i)$ as a Morse function. What we're doing is we're providing the ways to "fill" the gabs to create a diffeomorphism.

4 The Space of Objects

Theorem 4.1

Assuming the Cobordism Hypothesis and the PL-*Schoenflies Conjecture (in dimension* 4*),,*

Lemma 4.2

Assuming the PL-*Schoenflies Conjecture (in dimension* 4*), the space of objects of* $\mathcal{B}\mathrm{ord}^n$ *is* $\mathrm{PL}(n)/\mathrm{O}(n)$*.*

Lemma 4.3

Assuming the PL-*Schoenflies Conjecture (in dimension* 4*), the space of objects of* $\mathcal{B}\mathrm{ord}^n_{\mathrm{PL}}$ *is contractible.*

Proof: We can identify the space of objects with the space of controlled PL-homeomorphisms of $\mathbb{R}^n$, which we denote X. We show that the map $\mathrm{PL}_\partial(D^n) \to X$ is a trivial Kan fibration. ■

References